

DATA ANALYTICS

Data analytics is the process of examining raw data sets using specialized tools and techniques to uncover hidden patterns, trends, and correlations, transforming them into actionable insights. It enables businesses to make data-driven decisions, optimize efficiency, and predict future trends, ranging from basic descriptive reporting to advanced predictive modeling.

Features:

- Helps identify trends, patterns and anomalies in data to make informed decisions.
- Uses historical and current data to forecast future outcomes and opportunities.
- Involves building systems and tools to efficiently handle, process and analyse large volumes of data.

STEPS/LIFECYCLE

The data analytics lifecycle encompasses a series of stages that data analysts follow to turn raw data into meaningful insights. This process typically includes problem definition, data collection, data cleaning, data exploration, data transformation, data visualization, interpretation of results, and deployment of solutions. Each stage is integral to ensuring that the analysis is thorough and effective.

- **Problem Definition:** Forms the basis for the entire analytical process.
- **Data Collection:** Gathers relevant data, ensuring that the analysis is based on accurate information.
- **Data Cleaning:** Refines the dataset, eliminating errors that could skew results.
- **Data Exploration:** Identifies trends and patterns, providing insights that might not be immediately apparent.
- **Data Transformation:** This step enhances data quality and compatibility, typically involving steps like cleansing, mapping, and aggregation to ensure data is accurate and standardized.
- **Data Visualization:** It enables identifying patterns, trends, and outliers, facilitating faster and smarter decision-making. Key types include bar charts, scatter plots, and heatmaps
- **Interpretation of Results:** Helps stakeholders understand the implications of the findings.
- **Deployment:** Implements insights in real-world scenarios to drive business decisions.

DATA ANALYSIS TOOLS

Data analysis tools are software programs, applications, and other aids that professionals use to analyse data sets in ways that characterise the big picture of the information and provide usable information for insights, predictions, and decision-making purposes.

You can describe them as a collection of maps, diagrams, and charts designed to gather, interpret, and visualise data across diverse applications. Choosing the right data analysis tool is essential to maximise your efforts and make it easier to meet your professional goals.

Data Analysis Tool	Type	Ideal For	Best For	Pricing
Microsoft Excel	Spreadsheet	Business Analysts, Managers	Basic data manipulation	Paid (Microsoft 365)
Google Sheets	Spreadsheet	Individuals, Small-Medium Businesses	Basic data analysis and collaboration	Free with Paid upgrades
Google Analytics	Web Analytics	Digital Marketers, Web Analysts	Digital marketing analysis	Free and Paid (Google Analytics 360)
RapidMiner	Data Science	Data Scientists, Analysts	Predictive analytics	Free and Paid (various licensing options)
Tableau	Data Visualization	Business Analysts, Data Teams	Interactive dashboards	Paid (various plans)

Power BI	Business Intelligence	Business Analysts, Enterprises	Business reporting	Paid (various plans)
KNIME	Visual Workflow	Data Scientists, Analysts	Data science workflows	Free and Open source
Zoho Analytics	Business Intelligence	Small-Medium Businesses	Collaboration and reporting	Paid (various plans)
Qlik Sense	Business Intelligence	Business Analysts, Data Teams	Interactive analysis	Paid (various plans)

1.MS Excel

The world's best and most user-friendly spreadsheet software features calculations and graphing functions—and it's ideal for non-techies to perform basic data analysis and create charts and reports.

Pros

- No coding is required.
- User-friendly interface.

Cons

- Runs slow with complex data analysis.
- Less automation compared to specialized tools.

2.Google sheets

Similar to Microsoft Excel, Google Sheets stands out as a remarkable and cost-effective tool for fundamental data analysis. Not only does it handle everyday data analysis tasks, including sorting, filtering, and simple calculations, but it is known for its seamless collaboration capabilities as well.

Pros

- Easily accessible.
- Compatible with Microsoft Excel.
- Seamless integration with other Google Workspace tools.

Cons

- Lacks advanced features such as in Microsoft Excel.
- May not be able to handle large datasets.

3. Google Analytics

Digital marketers and web analysts use this one widely, and this tool helps businesses understand how people interact with their websites and apps, and it provides insights into website traffic, user behaviour, and performance to make **data-driven business decisions**.

Pros

- Free version available.
- Integrates with Google services.

Cons

- Limited customization for specific business needs.
- May not support non-web data sources.

4. RapidMiner

RapidMiner is ideal for data mining and model development, and it's a platform that offers remarkable machine learning and predictive analytics capabilities. It allows professionals to work with data at many stages, including preparation, information visualization, and analysis.

Pros

- User-friendly interface.
- Excellent support for machine learning.
- Large library of pre-built models.

Cons

- Can be expensive for advanced features.
- Limited data integration capabilities.

5. Tableau

Tableau is one of the best commercial data analysis tools you can get, and it's famous for its interactive dashboards and data exploration capabilities. Data teams can create visually appealing and interactive data representations through its easy-to-use interface and powerful capabilities.

Pros

- Intuitive drag-and-drop interface.
- Interactive and dynamic data visualization.
- Backed by Salesforce.

Cons

- Steeper learning curve for advanced features.

6. Power BI

This one is an excellent choice for creating insightful business dashboards, and among its nifty features it boasts incredible data integration features and interactive reporting, so making it ideal for enterprises.

Pros

- Intuitive drag-and-drop interface.
- Interactive and dynamic data visualization.
- Backed by Salesforce.

Cons

- Steeper learning curve for advanced features.

7. KNIME

The name is short for Konstanz Information Miner, and KNIME is an outstanding tool for data mining, with its user-friendly graphical interface that makes it accessible even to non-technical users, enabling them to create data workflows easily. What's more, KNIME is a cost-effective choice, and hence, it is ideal for small businesses operating on a limited budget.

Pros

- Visual workflow for data blending and automation.
- Active community and user support.

Cons

- Complex for beginners.
- Limited real-time data processing.

8. Zoho Analytics

Artificial intelligence and machine learning fuel this one, and Zoho Analytics is a robust data analysis platform. A neat point is how its data integration capabilities empower you to seamlessly connect and import data from diverse sources while offering an extensive array of analytical functions.

Pros

- Affordable pricing options.
- User-friendly interface.

Cons

- Limited scalability for very large datasets.
- Not as widely adopted as some other tools.

9. Qlik Sense

Qlik Sense offers you a wide range of augmented capabilities, and it's got everything from AI-generated analysis and insights to automated creation and data prep, machine learning, and predictive analytics.

Pros

- Impressive data exploration and visualization features.
- Can handle large datasets.

Cons

- Steep learning curve for new users.